DATA ANALYST - CONVEXITY SENSITIVITY AI AGENT

Category: Quantitative Finance | Fixed Income Analytics | Machine Learning

Timeline: 15 Days | AI-Accelerated Innovation & Research Project

Target: Finance / Engineering Students & Quant Analysts

TASK:

Build an AI Agent that analyses bond portfolio duration and convexity sensitivity under changing market conditions using Monte Carlo simulations, machine learning models (Random Forest, XGBoost, Neural Networks), and yield curve analytics. Implement in Python, R, and Excel/Power BI with advanced DAX. Design a gamified Bond Risk Lab simulation platform for training fixed income risk analysts.

PART A: TRAINING MATERIAL & LEARNING RESOURCES

PART B: GAMIFIED SIMULATION BUSINESS CONCEPT

PART C: CASE STUDIES & REAL-WORLD APPLICATIONS

PART D: 15-DAY PROJECT METHODOLOGY & DELIVERABLES

PART E: NO-CODE TOOLS & PLATFORMS GUIDE

PART F: SUBMISSION PROTOCOL

Important Note:

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EXECUTIVE SUMMARY

This project positions students as Lead Quantitative Analysts at a fictional Indian asset management company – Pinnacle Fixed Income Partners – managing a multi-currency bond portfolio worth INR 3,200+ crore across 300 bonds spanning Indian Government Securities, State Development Loans, Corporate Bonds (AAA to A rated), and International Sovereigns (US Treasuries, UK Gilts, German Bunds, JGBs, Australian Government Bonds).

The core challenge students must solve: The RBI has signalled a shift in monetary policy stance, and the Chief Investment Officer needs an AI-powered system to stress-test the entire portfolio under 1,000 interest rate scenarios within 24 hours. How does a portfolio manager quantify the exact INR impact of a 100 basis point parallel shift, a yield curve steepening, or a butterfly movement – and how can machine learning accelerate this analysis beyond what classical bond mathematics alone can deliver?

Students will design Bond Risk Lab – a gamified simulation platform that trains fixed income risk analysts through realistic, scenario-driven daily challenges. The platform spans all critical competencies: Macaulay Duration, Modified Duration, Effective Duration, Convexity, Key Rate Duration, DV01/PVBP, Monte Carlo VaR, Expected Shortfall, yield curve modelling (Nelson-Siegel, Svensson), and ML-based bond price prediction using Random Forest, XGBoost, and Deep Neural Networks.

The global fixed income market exceeds $130 trillion, with India's government securities market alone at INR 90+ lakh crore. Duration and convexity management is the fundamental skill for Treasury desks, ALM teams, pension funds, and insurance companies – yet training for this quantitative discipline remains largely textbook-based. Bond Risk Lab addresses this gap through gamification, instant computational feedback, and real-time scenario generation.

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PART A: TRAINING MATERIAL & LEARNING RESOURCES

SECTION A1: DOMAIN FUNDAMENTALS – THE FIXED INCOME RISK ECOSYSTEM

A1.1 Introduction To Duration And Convexity

Duration and convexity are the two most important risk measures in fixed income portfolio management. They quantify how sensitive a bond's price is to changes in interest rates – the dominant risk factor for all fixed income instruments. A portfolio manager who does not understand duration and convexity is essentially flying blind when interest rates move.

Definition And Core Concept:

Duration: The weighted average time to receive a bond's cash flows, expressed in years. It measures the first-order (linear) sensitivity of bond price to yield changes. A bond with duration of 7 years will lose approximately 7% of its value for every 1% increase in yield.

Convexity: The curvature of the price-yield relationship. It measures the second-order (quadratic) sensitivity of bond price to yield changes. Convexity provides a correction to the linear duration approximation, which becomes increasingly inaccurate for large yield movements (>50 basis points).

Why Both Measures Are Essential:

- Duration alone underestimates price gains when yields fall and overestimates losses when yields rise
- Positive convexity is desirable: a bond gains more when yields fall than it loses when yields rise by the same amount
- For large yield movements (100+ bps), the convexity adjustment can represent 10-30% of the total price change
- Callable bonds exhibit negative convexity at low yields – their price is capped by the call price
- Portfolio managers trade convexity as a distinct risk factor, separate from duration

The Price Change Approximation:

$$\Delta P/P \approx -\text{Modified Duration} \times \Delta y + \tfrac{1}{2} \times \text{Convexity} \times (\Delta y)^2$$

Where ΔP is the price change, P is the current price, and Δy is the yield change in decimal form.

Example – Why Convexity Matters:

Consider a 10-year bond with Modified Duration = 8.5 and Convexity = 95. If yields rise by 200 basis points (0.02):

Duration effect: $-8.5 \times 0.02 = -17.00\%$ (predicts 17% price decline)

Convexity correction: $+\tfrac{1}{2} \times 95 \times (0.02)^2 = +1.90\%$

Net price change: $-17.00\% + 1.90\% = -15.10\%$

Without the convexity correction, the duration-only estimate would overstate the loss by 190 basis points – a material error for a large portfolio.

A1.2 Types Of Duration

Macaulay Duration:

Named after Frederick Macaulay (1938), this is the weighted average time to receive all cash flows from a bond, where weights are the present values of each cash flow as a proportion of the bond's total price.

$$\text{MacD} = \Sigma_{t=1 \text{ to } n} [t \times PV(CF_t)] / \text{Price}$$

Python Implementation:

```
def macaulay_duration(face, coupon_rate, ytm, years, freq=2):
    c = face * coupon_rate / freq
    n = int(years * freq)
    r = ytm / freq
    price = bond_price(face, coupon_rate, ytm, years, freq)
    duration = 0
    for t in range(1, n+1):
        cf = c if t < n else c + face
        pv = cf / (1+r)**t
        duration += (t / freq) * pv
    return duration / price
```

Modified Duration:

Adjusts Macaulay Duration for the compounding effect, providing a direct measure of percentage price sensitivity per unit yield change:

```
ModD = MacD / (1 + y/freq)
```

Effective Duration (For Bonds With Embedded Options):

Uses finite difference method to capture the effect of optionality (callable/putable bonds):

```
EffD = (P_down - P_up) / (2 × P_0 × Δy)
where P_down = price if yield decreases by Δy
      P_up = price if yield increases by Δy
```

Key Rate Duration:

Decomposes interest rate sensitivity across specific maturity points on the yield curve. Instead of a single duration number, KRD provides a vector of sensitivities showing how the portfolio responds to movements at each tenor:

```
KRD_i = -(P_bumped_i - P_base) / (P_base × bump_size)
where only tenor point i is bumped, all others held constant
```

3M	6M	1Y	2Y	3Y	5Y	7Y	10Y	20Y	30Y
0.02	0.05	0.12	0.35	0.48	0.92	1.15	1.85	0.65	0.21

Sum of KRDs approximates Modified Duration (5.80 in this example).

DV01 / PVBP (Price Value of a Basis Point):

The absolute change in bond price for a 1 basis point (0.01%) change in yield. For a portfolio, DV01 is the most operationally useful risk measure:

```
DV01 = Modified Duration × Price / 10,000
Portfolio DV01 = Σ (DV01_i × Quantity_i)
```

A1.3 Convexity – Comprehensive Treatment

Mathematical Definition:

```
Convexity = (1/P) × d²P/dy² = Σ[t(t+1) × PV(CF_t)] / [P × freq² × (1+y/freq)²]
```

Python Implementation:

```
def convexity(face, coupon_rate, ytm, years, freq=2):
    c = face * coupon_rate / freq
    n = int(years * freq)
    r = ytm / freq
    price = bond_price(face, coupon_rate, ytm, years, freq)
    conv = 0
    for t in range(1, n+1):
        cf = c if t < n else c + face
        pv = cf / (1+r)**t
        conv += t * (t+1) * pv
    return conv / (price * freq**2 * (1+r)**2)
```

Properties Of Convexity:

- Zero-coupon bonds have the highest convexity for a given maturity (all cash flow at end)
- Higher maturity → higher convexity (exponential relationship)
- Lower coupon → higher convexity (cash flows more concentrated at maturity)
- Lower yield → higher convexity (discount factors amplify distant cash flows)
- Convexity is always positive for non-callable bonds (price-yield curve is always concave upward)
- Callable bonds: convexity turns negative as yields fall below the coupon rate (price compression toward call price)

Dollar Convexity:

```
Dollar Convexity = Convexity × Price
Dollar Price Change from Convexity = ½ × Dollar Convexity × (Δy)²
```

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Convexity Of Common Indian Instruments:

Instrument	Maturity	ModDur	Convexity	Convexity Impact @100bps
91-day T-Bill	0.25 yrs	0.24	0.08	+0.0004%
364-day T-Bill	1.0 yr	0.97	1.40	+0.007%
7.26% GOI 2033	9.0 yrs	6.85	58.20	+0.291%
7.54% GOI 2036	12.0 yrs	8.42	89.50	+0.448%
6.10% GOI 2054	30.0 yrs	14.20	285.00	+1.425%
AAA Corp Bond 3Y	3.0 yrs	2.72	10.50	+0.053%
AA Corp Bond 5Y	5.0 yrs	4.25	23.80	+0.119%

A1.4 Yield Curve Dynamics And Risk Scenarios

Yield Curve Shapes And Economic Interpretation:

- Normal (upward sloping): Short rates below long rates; economy in expansion; RBI policy rate below neutral
- Flat: Short rates equal long rates; economy transitioning; uncertainty about future rates
- Inverted: Short rates above long rates; economy heading toward recession; tight monetary policy
- Humped: Middle maturities higher than both short and long; mixed market expectations

Three Fundamental Yield Curve Movements:

1. Parallel Shift (Level):

The entire yield curve moves up or down by the same amount. This is the simplest scenario and is fully captured by Modified Duration. Example: RBI surprises with a 50bps rate cut – all yields drop approximately 50bps.

$$\text{Portfolio P\&L} = -\text{Portfolio ModDur} \times \Delta y \times \text{Portfolio Market Value} + 0.5 \times \text{Portfolio Convexity} \times (\Delta y)^2 \times \text{Portfolio MV}$$

2. Twist / Steepening-Flattening (Slope):

Short end and long end move differently. The 2s10s spread (10Y yield minus 2Y yield) widens (steepening) or narrows (flattening). Key Rate Duration is required to capture this – a single duration number is insufficient.

```
P&L = Σ(KRD_i × Δy_i × Portfolio MV_i) for each tenor bucket i
```

3. Butterfly (Curvature):

The belly of the curve (5Y-7Y) moves differently from both the short end and the long end. This creates a hump or trough in the curve. Portfolio impact depends on the maturity distribution of holdings.

```
Butterfly P&L depends on distribution of holdings across the curve
A barbell portfolio (concentrated in short + long) behaves differently
from a bullet portfolio (concentrated in the belly)
```

RBI Monetary Policy And Indian Yield Curve:

- Repo Rate: The RBI's primary policy tool; currently anchors the short end of the INR yield curve
- LAF Corridor: Reverse Repo to MSF rate; defines the overnight rate band
- G-Sec Benchmark: The 10-year benchmark G-Sec (currently 7.26% 2033) is the reference for long-end pricing
- Operation Twist: RBI simultaneously buys long-dated and sells short-dated G-Secs to flatten the curve
- SDL Spread: State Development Loans trade at a spread over G-Secs; spread varies by state credit quality

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SECTION A2: MONTE CARLO SIMULATION AND MACHINE LEARNING

A2.1 Monte Carlo Simulation For Interest Rate Scenarios

Monte Carlo simulation generates thousands of possible future yield curve states by sampling from probability distributions calibrated to historical data. For each scenario, the portfolio is reevaluated using duration-convexity approximation (fast) or full repricing (accurate), generating a distribution of possible P&L outcomes.

Complete Python Monte Carlo Engine:

```
import numpy as np
from scipy.stats import norm

class BondRiskMC:
    def __init__(self, portfolio_df, n_sims=1000):
        self.portfolio = portfolio_df
        self.n = n_sims
        self.pnl = []

    def generate_scenarios(self, vol_p=0.01, vol_t=0.005,
vol_b=0.003):
        np.random.seed(42)
        self.parallel = np.random.normal(0, vol_p, self.n)
        self.twist = np.random.normal(0, vol_t, self.n)
        self.butterfly = np.random.normal(0, vol_b, self.n)

    def compute_pnl(self):
        total_mv = self.portfolio['MarketValue_INR'].sum()
        port_dur = (self.portfolio['ModifiedDuration']
            * self.portfolio['PortfolioWeight']).sum()
        port_conv = (self.portfolio['Convexity']
            * self.portfolio['PortfolioWeight']).sum()
        for i in range(self.n):
            dy = self.parallel[i]
            dur_eff = -port_dur * dy * total_mv
            conv_eff = 0.5 * port_conv * dy**2 * total_mv
            self.pnl.append(dur_eff + conv_eff)

    def var(self, conf=0.95):
        return np.percentile(self.pnl, (1-conf)*100)

    def cvar(self, conf=0.95):
```

```
        v = self.var(conf)
        return np.mean([x for x in self.pnl if x <= v])
```

A2.2 Machine Learning For Bond Price Prediction

Random Forest Regressor (Python):

```
from sklearn.ensemble import RandomForestRegressor
X =
df[['coupon_rate','maturity','yield_rate','duration','convexity']]
y = df['price']
model = RandomForestRegressor(n_estimators=100, random_state=42)
model.fit(X_train, y_train)
rmse = np.sqrt(mean_squared_error(y_test, model.predict(X_test)))
```

XGBoost Regressor:

```
from xgboost import XGBRegressor
xgb = XGBRegressor(n_estimators=200, max_depth=4,
learning_rate=0.1)
xgb.fit(X_train, y_train)
```

Neural Network (TensorFlow):

```
model = tf.keras.Sequential([
    tf.keras.layers.Dense(128, activation='relu'),
    tf.keras.layers.Dropout(0.2),
    tf.keras.layers.Dense(64, activation='relu'),
    tf.keras.layers.Dense(1)])
```

SHAP Explainability:

```
import shap
explainer = shap.TreeExplainer(model)
shap_values = explainer.shap_values(X_test)
shap.summary_plot(shap_values, X_test)
```

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A2.3 R Implementation

```
library(tidyverse); library(caret); library(xgboost); library(h2o)
gbm_dur <- train(ModifiedDuration ~ CouponRate + YearsToMaturity +
YieldToMaturity,
    data=train, method='gbm',
trControl=trainControl(method='cv',number=5))
xgb_conv <- xgboost(data=as.matrix(train_x), label=train_y,
    max_depth=4, eta=0.1, nrounds=100, verbose=0)
h2o.init()
nn <- h2o.deeplearning(x=features, y='Duration',
training_frame=h2o_data,
    activation='RectifierWithDropout', hidden=c(100,50),
epochs=50)
```

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SECTION A3: YIELD CURVE MODELLING

A3.1 Nelson-Siegel Model

The Nelson-Siegel model parameterises the yield curve using four parameters that control level (β0), slope (β1), curvature (β2), and decay rate (λ):

```
y(τ) = β0 + β1 × [(1-exp(-τ/λ))/(τ/λ)] + β2 × [(1-exp(-τ/λ))/(τ/λ) - exp(-τ/λ)]
```

```
from scipy.optimize import minimize
def nelson_siegel(tau, b0, b1, b2, lam):
    x = tau / lam
    f1 = (1 - np.exp(-x)) / x
    f2 = f1 - np.exp(-x)
    return b0 + b1*f1 + b2*f2
```

A3.2 Svensson Extension

Adds two more parameters (β3, λ2) for a second hump, providing better fit for complex shapes:

```
y(τ) = NS(τ) + β3 × [(1-exp(-τ/λ2))/(τ/λ2) - exp(-τ/λ2)]
```

A3.3 Bootstrapping Zero-Coupon Curve

Bootstrap method derives spot rates from observed coupon bond prices iteratively from shortest to longest maturity. Each new spot rate is solved using all previously derived rates:

```
P = C/(1+z1) + C/(1+z2)^2 + ... + (C+F)/(1+z_n)^n
Solve for z_n given known z_1 through z_(n-1)
```

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SECTION A4: ADVANCED DAX FOR FIXED INCOME DASHBOARDS

A4.1 Portfolio-Level Measures

```
Portfolio ModDur = SUMX(Bonds, Bonds[ModifiedDuration] *
Bonds[PortfolioWeight])

Portfolio Convexity = SUMX(Bonds, Bonds[Convexity] *
Bonds[PortfolioWeight])

Portfolio DV01 = SUMX(Bonds, Bonds[DV01_Per100Face] *
Bonds[Quantity] / 100)

Price Impact =
    VAR dy = SELECTEDVALUE('Yield Change'[Value], 0)
    VAR DurEff = -[Portfolio ModDur] * dy * [Total MV]
    VAR ConvEff = 0.5 * [Portfolio Convexity] * dy^2 * [Total MV]
    RETURN DurEff + ConvEff
```

A4.2 VaR and Risk Measures

```
VaR 95% = PERCENTILEX.INC(MCScenarios, MCScenarios[PnL_Total_INR],
0.05)

CVaR 95% = AVERAGEX(FILTER(MCScenarios,
MCScenarios[PnL_Total_INR]<=[VaR 95%]),
    MCScenarios[PnL_Total_INR])

Duration Contribution by Rating =
    SUMX(FILTER(Bonds,
Bonds[CreditRating]=SELECTEDVALUE(Bonds[CreditRating])),
        Bonds[ModifiedDuration] * Bonds[PortfolioWeight])

KRD by Bucket =
    SUMX(FILTER(Bonds,
Bonds[KeyRateBucket]=SELECTEDVALUE(DimTenor[Bucket])),
        Bonds[ModifiedDuration] * Bonds[PortfolioWeight])
```

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A4.3 Excel VBA Implementation

```
Function BondPrice(Face, Coupon, YTM, Years) As Double
    Dim c As Double: c = Face * Coupon / 2
    Dim n As Integer: n = Years * 2
    Dim r As Double: r = YTM / 2
    BondPrice = c * (1-(1+r)^-n)/r + Face/(1+r)^n
End Function

Function MacaulayDur(Face, Coupon, YTM, Years) As Double
    Dim dur As Double: dur = 0
    Dim P As Double: P = BondPrice(Face, Coupon, YTM, Years)
    For t = 1 To Years * 2
        If t < Years*2 Then cf = Face*Coupon/2 Else cf =
Face*Coupon/2+Face
        dur = dur + (t/2) * cf / (1+YTM/2)^t
    Next t
    MacaulayDur = dur / P
End Function
```

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SECTION A4.1: COMPREHENSIVE WORKED EXAMPLES

Worked Example 1: Single Bond – 7.26% Government Of India 2033

Given:

- Face Value: INR 100
- Coupon Rate: 7.26% (semi-annual)
- Yield to Maturity: 7.35%
- Years to Maturity: 9.0
- Coupon Frequency: 2 (semi-annual)

Step 1: Bond Price

```
Semi-annual coupon = 100 × 0.0726 / 2 = 3.63
Number of periods = 9 × 2 = 18
Semi-annual yield = 0.0735 / 2 = 0.03675
PV of coupons = 3.63 × [1 - (1.03675)^-18] / 0.03675
              = 3.63 × 13.0567 = 47.4160
PV of principal = 100 / (1.03675)^18 = 52.0021
Clean Price = 47.4160 + 52.0021 = 99.4181
```

Step 2: Macaulay Duration

```
Sum of (t/2) × PV(CF_t) for t = 1 to 18:
Period 1: (0.5) × 3.63/(1.03675)^1  = 0.5 × 3.5015 = 1.7508
Period 2: (1.0) × 3.63/(1.03675)^2  = 1.0 × 3.3768 = 3.3768
...
Period 18: (9.0) × 103.63/(1.03675)^18 = 9.0 × 53.8881 = 484.9929
Sum = 684.32 (all periods)
Macaulay Duration = 684.32 / 99.4181 = 6.8831 years
```

Step 3: Modified Duration

```
Modified Duration = 6.8831 / (1 + 0.0735/2) = 6.8831 / 1.03675 =
6.6389
```

Step 4: Convexity

```
Sum of t(t+1) × PV(CF_t) for t = 1 to 18:
Convexity = Sum / (Price × freq² × (1+y/freq)²)
          = 26,546.80 / (99.4181 × 4 × 1.07431)
          = 62.16
```

Step 5: DV01

```
DV01 = 6.6389 × 99.4181 / 10,000 = 0.0660 per 100 face
```

Step 6: Price Impact Estimation

If yields rise by 100 basis points (0.01):

```
Duration effect: -6.6389 × 0.01 × 99.4181 = -6.6007
Convexity effect: +0.5 × 62.16 × (0.01)² × 99.4181 = +0.3089
Estimated new price = 99.4181 - 6.6007 + 0.3089 = 93.1263
Actual new price (repriced at 8.35%): 93.16 (close match!)
```

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Worked Example 2: Portfolio – 3-Bond Portfolio

Portfolio:

Bond	ModDur	Convexity	DV01	Price	Quantity	MV (INR)
GOI 2033	6.64	62.2	0.066	99.42	5,000	4,97,100
AAA Corp 2027	2.72	10.5	0.027	101.50	3,000	3,04,500
GOI 2054	14.20	285.0	0.143	95.80	2,000	1,91,600

```
Total MV = 4,97,100 + 3,04,500 + 1,91,600 = 9,93,200

Weights: GOI 2033 = 50.05%, AAA Corp = 30.66%, GOI 2054 = 19.29%

Portfolio ModDur = 6.64×0.5005 + 2.72×0.3066 + 14.20×0.1929
                 = 3.3233 + 0.8340 + 2.7392 = 6.8965

Portfolio Convexity = 62.2×0.5005 + 10.5×0.3066 + 285.0×0.1929
                    = 31.13 + 3.22 + 54.98 = 89.33

Portfolio DV01 = 0.066×50 + 0.027×30 + 0.143×20 = 3.30+0.81+2.86 =
6.97
```

Scenario: Parallel shift +100bps

```
Duration P&L = -6.8965 × 0.01 × 993,200 = -INR 68,501
Convexity P&L = +0.5 × 89.33 × (0.01)² × 993,200 = +INR 4,435
Net P&L = -68,501 + 4,435 = -INR 64,066 (-6.45%)

Without convexity: -INR 68,501 (-6.90%)
Convexity saved: INR 4,435 (0.45% of portfolio)
```

Scenario: Non-parallel steepening (2Y +80bps, 10Y +50bps, 30Y +30bps)

```
AAA Corp (2Y bucket): -2.72 × 0.008 × 304,500 = -INR 6,626
GOI 2033 (10Y bucket): -6.64 × 0.005 × 497,100 = -INR 16,504
GOI 2054 (30Y bucket): -14.20 × 0.003 × 191,600 = -INR 8,162
Total non-parallel P&L = -INR 31,292
(vs parallel +50bps would give -INR 34,250 – different!)
```

Worked Example 3: VaR Calculation

Given: Portfolio MV = INR 9,93,200, ModDur = 6.90, Convexity = 89.33

```
1-day yield volatility: 6 basis points (0.0006)

Parametric VaR 95% (1-day):
  = MV × ModDur × z_95% × daily_vol
  = 993,200 × 6.90 × 1.645 × 0.0006
  = 993,200 × 6.90 × 0.000987
  = INR 6,764

Parametric VaR 99% (1-day):
  = 993,200 × 6.90 × 2.326 × 0.0006
  = INR 9,568

10-day VaR (for regulatory capital):
  = 1-day VaR × √10
  = 6,764 × 3.162 = INR 21,387 (95%)
  = 9,568 × 3.162 = INR 30,244 (99%)
```

In practice, Monte Carlo VaR with convexity adjustment will produce slightly different (typically lower) VaR numbers because convexity partially cushions losses. The difference increases with portfolio convexity and yield volatility.

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SECTION A5: BOND PORTFOLIO DATA REQUIREMENTS

A5.1 Data Structure For Duration-Convexity Analysis

The AI agent requires granular bond-level data to compute accurate duration, convexity, and risk measures. The following data elements must be sourced from the bank's Treasury Management System, Bloomberg terminals, or SAP/ERP systems. Each field is critical for at least one calculation in the risk framework.

Bond Identification Fields:

Field Name	Data Type	Description	Calculation Use
BondID	VARCHAR(15)	Internal bond identifier	Tracking, aggregation
ISIN	VARCHAR(12)	International Securities Identification Number	Market data linkage
Issuer	VARCHAR(100)	Bond issuer name	Concentration analysis
Sector	VARCHAR(30)	Government/Banking/NBFC/Corporate	Sector risk attribution
CreditRating	VARCHAR(5)	Current credit rating (SOV/AAA/AA/A/BBB)	Risk-weighted duration
Currency	VARCHAR(3)	Denomination currency (INR/USD/GBP/EUR/JPY)	Multi-currency VaR

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Bond Terms And Cash Flow Fields:

Field Name	Data Type	Description	Calculation Use
FaceValue	DECIMAL(15,2)	Par value of bond	Cash flow base
CouponRate	DECIMAL(6,4)	Annual coupon rate (decimal)	Duration, convexity input
CouponFrequency	INT	Payments per year (1=annual, 2=semi)	Discounting frequency
BondType	VARCHAR(10)	Fixed/Floating/Zero-Coupon	Valuation method
IssueDate	DATE	Original issue date	Accrued interest calc
MaturityDate	DATE	Final maturity date	Duration, convexity input
YearsToMaturity	DECIMAL(8,4)	Years remaining to maturity	Primary duration input
IsCallable	BOOLEAN	Bond has embedded call option	Effective duration required
CallDate	DATE	First call date (if callable)	Option-adjusted analysis

Market Data And Valuation Fields:

Field Name	Data Type	Description	Calculation Use
YieldToMaturity	DECIMAL(6,4)	Current YTM (decimal)	Primary pricing input
CleanPrice	DECIMAL(10,4)	Market price ex-accrued	Portfolio valuation
AccruedInterest	DECIMAL(10,4)	Accrued coupon to date	Dirty price = Clean + AI
DirtyPrice	DECIMAL(10,4)	Settlement price (Clean + AI)	Actual market value
SpreadOverBenchmark	INT	Credit spread in basis points	OAS, Z-spread analysis
BenchmarkIndex	VARCHAR(10)	Reference benchmark (IN10YT, UST10Y)	Relative value

Computed Risk Metrics (Calculated By AI Agent):

Field Name	Data Type	Description	Calculation Use
MacaulayDuration	DECIMAL(8,4)	Weighted avg time to cash flows	SEBI scheme classification
ModifiedDuration	DECIMAL(8,4)	% price sensitivity per 1% yield change	Primary risk measure
EffectiveDuration	DECIMAL(8,4)	Duration adjusted for optionality	Callable/putable bonds
Convexity	DECIMAL(10, 4)	Second-order price sensitivity	Large move correction
DV01	DECIMAL(10, 4)	Price change per 1 basis point	Hedging, risk limits
KeyRateBucket	VARCHAR(10)	Maturity bucket for KRD	Curve risk decomposition
PortfolioWeight	DECIMAL(8,6)	Market value weight in portfolio	Weighted aggregation

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A5.2 Yield Curve Data Requirements

The AI agent requires historical yield curve data to calibrate Monte Carlo simulations and fit parametric models. The yield curve data table provides spot yields at each tenor point:

Field Name	Data Type	Description	Example
CurveDate	DATE	Observation date	2024-10-31
Currency	VARCHAR(3)	Yield curve currency	INR
Tenor_Years	DECIMAL(5,2)	Maturity point in years	10.00
Yield	DECIMAL(8,6)	Spot yield at tenor	0.073500
ZeroRate	DECIMAL(8,6)	Zero-coupon rate (bootstrapped)	0.074200
ForwardRate	DECIMAL(8,6)	Implied forward rate	0.078100
DiscountFactor	DECIMAL(8,6)	PV of 1 unit at tenor	0.495362

Standard INR Yield Curve Tenors:

The following 11 tenor points are standard for Indian fixed income analytics and must be included in all yield curve data: 3 months (91-day T-Bill), 6 months (182-day T-Bill), 1 year (364-day T-Bill), 2 years, 3 years, 5 years, 7 years, 10 years (benchmark G-Sec), 15 years, 20 years, and 30 years (ultra-long G-Sec).

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A5.3 Monte Carlo Scenario Data Structure

Field Name	Data Type	Description	Example
ScenarioID	INT	Unique scenario identifier	1 to 1000
ParallelShift_bps	DECIMAL(8,2)	Level factor in basis points	-47.25
TwistFactor_bps	DECIMAL(8,2)	Slope factor in basis points	12.80
ButterflyFactor_bps	DECIMAL(8,2)	Curvature factor in basis points	-3.15
PnL_DurationEffect	DECIMAL(15,2)	P&L from duration (first order)	1,245,678.50
PnL_ConvexityEffect	DECIMAL(15,2)	P&L from convexity (second order)	35,892.40
PnL_Total	DECIMAL(15,2)	Total scenario P&L	1,281,570.90
PnL_Pct	DECIMAL(8,4)	P&L as % of portfolio value	3.9975

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SECTION A6: DETAILED SCENARIO ANALYSIS FRAMEWORK

A6.1 Parallel Shift Scenarios

Standard Regulatory Scenarios (RBI/SEBI mandated for banks):

Scenario	Shift (bps)	Duration P&L	Convexity P&L	Net P&L
Minor tightening	+25	-ModDur×0.0025×MV	+0.5×Conv×(0.0025)²×MV	Duration dominates
RBI hike	+50	-ModDur×0.005×MV	+0.5×Conv×(0.005)²×MV	Convexity ~2-3%
Stress test	+100	-ModDur×0.01×MV	+0.5×Conv×(0.01)²×MV	Convexity ~5-8%
Severe stress	+200	-ModDur×0.02×MV	+0.5×Conv×(0.02)²×MV	Convexity ~10-15%
Crisis	+300	-ModDur×0.03×MV	+0.5×Conv×(0.03)²×MV	Convexity ~20%+
Rally	-100	+ModDur×0.01×MV	+0.5×Conv×(0.01)²×MV	Both positive!

Note: For a yield decline, both duration and convexity effects are positive (price goes up, and convexity adds further upside). This asymmetry – gaining more on rallies than losing on sell-offs – is the fundamental value of positive convexity.

A6.2 Non-Parallel Shift Scenarios

Scenario: Yield Curve Steepening (+50bps at 2Y, +100bps at 10Y, +75bps at 30Y)

This scenario cannot be captured by a single duration number. Key Rate Duration is required:

```
P&L = KRD_2Y × (-0.005) × MV_2Y_bucket
    + KRD_5Y × (-0.0075) × MV_5Y_bucket
    + KRD_10Y × (-0.01) × MV_10Y_bucket
    + KRD_30Y × (-0.0075) × MV_30Y_bucket
    + Convexity corrections at each tenor
```

Scenario: Yield Curve Inversion (Short rates +100bps, Long rates -25bps)

A barbell portfolio (concentrated in short + long maturities) would suffer at the short end but gain at the long end, partially offsetting. A bullet portfolio (concentrated in 5-7Y) would have a moderate loss. This demonstrates why KRD decomposition is essential for understanding exposure to curve shape changes.

A6.3 Multi-Factor Stress Test Framework

The most realistic scenario combines all three factors simultaneously. The following framework is used by bank Treasury desks for ALCO reporting:

Standard Stress Test Matrix:

Scenario Name	Parallel	Twist	Butterfly	Description
Base Case	0	0	0	No change from current curve
RBI Hike (Expected)	+25	0	0	Gradual tightening
RBI Surprise Hike	+50	-20	+5	Front-end led, flattening
Global Risk-Off	+75	+30	-10	Steepening, risk aversion
Dovish Pivot	-75	-20	+5	Rate cut expectations
Credit Crisis	+100	+50	-20	Flight to quality, steep curve
Stagflation	+150	+25	+10	High rates across curve
Black Swan	+300	+75	-30	Extreme stress, 3-sigma event

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SECTION A7: VALUE-AT-RISK AND TAIL RISK ANALYSIS

A7.1 Three Approaches To VaR

1. Parametric (Variance-Covariance) VaR:

Assumes portfolio returns are normally distributed. VaR = Portfolio Value × z-score × σ × √t, where z-score is 1.645 for 95% confidence and 2.326 for 99% confidence.

```
parametric_var_95 = total_mv * 1.645 * portfolio_vol *
np.sqrt(1/252)
```

2. Historical Simulation VaR:

Uses actual historical yield changes to generate P&L distribution. No distributional assumption required. Apply each historical daily yield change to the current portfolio using duration-convexity approximation.

```
historical_pnl = []
for daily_change in historical_yield_changes[-252:]:  # last year
    pnl = -port_dur * daily_change * total_mv
    pnl += 0.5 * port_conv * daily_change**2 * total_mv
    historical_pnl.append(pnl)
hist_var_95 = np.percentile(historical_pnl, 5)
```

3. Monte Carlo VaR:

Generates random scenarios from a calibrated distribution (not limited to normal). The most flexible approach – can incorporate fat tails, skewness, and correlation between risk factors.

```
# Already implemented in Section A2 - BondRiskMC class
mc = BondRiskMC(portfolio, n_sims=10000)
mc.generate_scenarios(vol_p=0.01, vol_t=0.005, vol_b=0.003)
mc.compute_pnl()
mc_var_95 = mc.var(0.95)
mc_cvar_95 = mc.cvar(0.95)
```

A7.2 Expected Shortfall (CVaR)

Expected Shortfall (also called Conditional VaR or CVaR) is the average loss in the tail beyond VaR. It provides information about the magnitude of tail losses, not just the threshold. Basel III/IV requires banks to use Expected Shortfall rather than VaR for market risk capital calculations.

```
CVaR = E[Loss | Loss > VaR]
     = Average of all P&L values that exceed the VaR threshold
```

A7.3 Backtesting VaR

VaR models must be backtested by comparing actual daily P&L against the predicted VaR. The number of VaR breaches (actual loss exceeding VaR) should be consistent with the confidence level:

- 95% VaR: Expected ~13 breaches per year (5% of 252 trading days)
- 99% VaR: Expected ~2.5 breaches per year (1% of 252 trading days)
- Basel traffic light system: Green (0-4 breaches), Yellow (5-9), Red (10+) for 99% VaR over 250 days

```
# Kupiec POF test for VaR backtesting
breaches = sum(1 for actual, var in zip(actual_pnl, var_series) if
actual < var)
expected = int(0.05 * len(actual_pnl))
print(f'Actual breaches: {breaches}, Expected: {expected}')
```

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SECTION A8: DURATION-BASED HEDGING STRATEGIES

A8.1 Immunisation

Portfolio immunisation matches the duration of assets and liabilities so that the portfolio value is insensitive to small interest rate changes. For a pension fund with a liability duration of 12 years, the asset portfolio must also have a duration of 12 years.

```
Target: Asset Duration = Liability Duration
If Asset Duration > Liability Duration: Portfolio gains when rates fall, loses when rates rise
If Asset Duration < Liability Duration: Opposite exposure
```

A8.2 Hedging With Interest Rate Swaps

Interest rate swaps are the primary tool for adjusting portfolio duration:

- Receive-Fixed Swap: Increases portfolio duration (equivalent to buying a fixed-rate bond and selling a floating-rate bond)
- Pay-Fixed Swap: Decreases portfolio duration (equivalent to selling a fixed-rate bond and buying floating)

```
Swap Notional Required = -(Portfolio DV01 - Target DV01) / DV01_per_unit_swap
```

A8.3 Barbell Vs Bullet Vs Ladder

Three portfolio construction strategies with identical duration but different convexity:

- Barbell: Concentrate in short (2Y) and long (30Y) bonds. High convexity. Benefits from curve flattening.
- Bullet: Concentrate around the target duration (e.g., all 7Y). Low convexity. Neutral to curve shape.
- Ladder: Equal allocation across maturities. Moderate convexity. Natural reinvestment diversification.

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Python Comparison:

```
# Compare convexity of three strategies with same ModDur = 7.0
barbell_conv = 0.3 * conv_2Y + 0.7 * conv_30Y  # High ~200+
bullet_conv  = 1.0 * conv_7Y                   # Low ~60
ladder_conv  = avg([conv_t for t in [2,5,7,10,15,20,30]])  #
Medium ~120

# Barbell outperforms in parallel shifts (convexity advantage)
# Bullet outperforms in curve steepening (no long-end exposure)
# Ladder provides stable reinvestment income
```


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PART B: GAMIFIED SIMULATION BUSINESS CONCEPT

Note on the Simulation Concept: The gamified simulation concept presented in this section is provided as a reference framework. Students may choose to follow this concept or design their own original gamified simulation - with a different name, structure, narrative, or game mechanics - as long as it meets the assessment criteria outlined in Part F.

SECTION B1: BOND RISK LAB – SIMULATION OVERVIEW

B1.1 Business Rationale And Problem Statement

Bond Risk Lab is a browser-based gamified simulation platform designed to train and certify fixed income risk analysts through realistic, time-pressured, scenario-driven challenges. The player assumes the role of a Fixed Income Risk Analyst at Pinnacle Fixed Income Partners, managing a portfolio of 300 bonds across 6 currencies and multiple credit ratings.

The Training Gap This Platform Solves:

Traditional fixed income training relies on textbook formulas and static Excel exercises. Trainees learn duration and convexity as mathematical concepts but never experience the real-time pressure of managing a live portfolio during an RBI policy announcement, a credit downgrade cascade, or a sudden yield curve inversion. The consequences of errors in live risk management are severe: incorrect hedging decisions based on wrong duration numbers can expose the firm to millions in unexpected losses.

India's fixed income market has grown to over INR 90 lakh crore in government securities alone. Every mutual fund, insurance company, pension fund, and bank Treasury requires skilled duration/convexity analysts – yet there is no standardised, scalable, engaging training platform. Bond Risk Lab fills this gap.

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Target Audience:

- Primary: Finance and engineering students preparing for Treasury, ALM, and risk management roles
- Secondary: Working professionals in fixed income trading and risk management seeking to upskill on advanced analytics
- Tertiary: CFA Level 1-2 candidates seeking practical fixed income problem-solving experience

B1.2 Simulation Architecture

Core Components:

- Portfolio Engine: Pre-built 300-bond portfolio across G-Secs, SDLs, Corporate Bonds, and International Sovereigns with real-time duration/convexity computation
- Yield Curve Simulator: Generates daily yield curve movements based on parameterised scenarios – normal markets, RBI surprise, credit crisis, curve inversion
- Monte Carlo Module: 1,000-scenario VaR engine that students must configure, run, and interpret
- ML Workbench: Pre-trained Random Forest, XGBoost, and Neural Network models for bond price prediction; students compare ML sensitivity vs analytical sensitivity
- Scenario Injector: Randomised daily events – RBI rate decision, credit downgrade, currency shock, liquidity crisis, curve twist
- Scoring Engine: Awards points for calculation accuracy, hedging decisions, risk reporting speed, and model validation quality
- Learning Layer: Every error triggers explanation with correct formula, Python code, and reference to Fabozzi/Hull textbook chapter

B1.3 Game Modes

Campaign Mode (Primary – 6 Levels):

- Level 1 – The Basics: Single G-Sec portfolio (10 bonds); compute duration and convexity; estimate price impact for ±50bps parallel shift
- Level 2 – Multi-Bond: 50-bond mixed portfolio (G-Secs + Corporate); DV01 computation; portfolio-level duration and convexity
- Level 3 – Yield Curve: Full 300-bond portfolio; non-parallel shifts introduced; Key Rate Duration decomposition required
- Level 4 – Monte Carlo: Run 1,000 scenarios; compute VaR/CVaR; identify which scenarios breach risk limits
- Level 5 – Machine Learning: Train RF/XGBoost models; compare ML sensitivity to analytical; SHAP explainability
- Level 6 – Crisis Mode: RBI emergency rate hike + credit downgrade + currency shock simultaneously; full stress test under time pressure

SECTION B2: SCORING SYSTEM AND GAME MECHANICS

B2.1 1000-Point Scoring Framework

Category 1 – Duration & Convexity Accuracy (250 Points):

- Full credit (250): All calculations correct to 4 decimal places vs Bloomberg benchmark
- Partial credit (125-249): Correct methodology, minor rounding differences
- Zero credit: Wrong formula applied (e.g., Macaulay used where Modified required)

Category 2 – Portfolio Risk Computation (200 Points):

- DV01 calculation accuracy: 80 points
- Key Rate Duration decomposition: 60 points
- Portfolio duration contribution by rating/sector/currency: 60 points

Category 3 – Monte Carlo & Var (200 Points):

- Correct scenario generation with appropriate volatility assumptions: 80 points
- VaR/CVaR calculation accuracy vs analytical benchmark: 80 points
- Risk limit breach identification: 40 points

Category 4 – Machine Learning Quality (150 Points):

- Model training with RMSE < 0.50: 60 points
- Sensitivity analysis using ML vs analytical comparison: 50 points
- SHAP/feature importance interpretation: 40 points

Category 5 – Yield Curve Modelling (100 Points):

- Nelson-Siegel fit quality ($R^2 > 0.99$): 50 points
- Correct identification of curve shape and economic interpretation: 50 points

Category 6 – Speed And Reporting (100 Points):

- All computations completed within time limit: 40 points
- Risk report formatted correctly: 30 points
- Hedging recommendation with rationale: 30 points

Penalties:

- Using Macaulay Duration where Modified required: –30 points per instance
- Ignoring convexity correction for moves >50bps: –20 points
- VaR computed at wrong confidence level: –40 points
- Missing yield curve scenario type in stress test: –25 points

B2.2 Progression System (7 Levels)

- Level 1 – Analyst Trainee: XP 0; Tutorial scenarios only; single bond calculations
- Level 2 – Junior Analyst: XP 500; Campaign Levels 1-2; portfolio aggregation
- Level 3 – Risk Analyst: XP 1,500; Campaign Levels 1-4; Monte Carlo introduced
- Level 4 – Senior Risk Analyst: XP 3,500; All Campaign Levels; ML models
- Level 5 – Portfolio Risk Manager: XP 7,000; Full crisis scenarios; hedging decisions
- Level 6 – Head of Fixed Income Risk: XP 15,000; Multi-currency; regulatory reporting
- Level 7 – Chief Risk Officer: XP 30,000; All scenarios mastered; can design challenges

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Achievement Badges (20 Badges):

- Duration Master: 50 consecutive perfect duration calculations
- Convexity Guru: Correctly identify negative convexity in callable bonds
- Monte Carlo Architect: Run 10,000-scenario simulation without errors
- ML Engineer: Train model with RMSE < 0.25
- VaR Champion: Correctly predict next-day P&L within VaR bounds for 10 days
- Nelson-Siegel Wizard: Fit yield curve with $R^2 > 0.999$
- Key Rate Ninja: Decompose KRD across 10 buckets with <1% total error
- Crisis Survivor: Complete Level 6 crisis with Distinction grade
- Speed Demon: Complete full risk report in under 15 minutes
- Hedging Hero: Recommend optimal hedge that reduces portfolio DV01 by 90%+

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SECTION B3: SCENARIO ENGINE AND BUSINESS MODEL

B3.1 Scenario Categories

Routine Scenarios (40%): Normal market day, yields move ±5-10bps, standard risk computation.

Yield Curve Scenarios (25%): Non-parallel movements – steepening, flattening, inversion, butterfly.

Credit Scenarios (15%): Rating downgrade, spread widening, flight to quality.

Crisis Scenarios (10%): RBI emergency action, simultaneous multi-factor shock.

ML Challenge Scenarios (10%): Predict price using ML model; compare to analytical; explain discrepancy.

B3.2 Sample Scenario: The RBI Surprise

It is 10:00 AM. The RBI announces an unexpected 50bps repo rate hike (market was expecting a hold). The yield curve immediately shifts: short end up 60bps, long end up 30bps (flattening). Your portfolio's modified duration is 5.8. What is the estimated P&L impact?

- Step 1: This is a non-parallel shift – you cannot use simple duration × Δy. You need KRD.
- Step 2: Apply KRD vector: short buckets × -60bps, long buckets × -30bps, interpolate middle
- Step 3: Add convexity correction (the shift is large enough to matter)
- Step 4: Report P&L, identify which holdings are worst affected, recommend hedge
- Bonus: Run Monte Carlo with updated volatilities to generate revised VaR

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B3.3 Business Model And Monetisation

Revenue Streams:

- B2B SaaS – Bank Treasury Training: INR 12,000-20,000/user/year; target 50 banks
- Insurance/Pension Fund Training: INR 15,000/user/year; target 25 insurers
- CFA/FRM Exam Preparation Module: INR 4,999/candidate; target 50,000 candidates/year
- University Licensing: INR 2-5 lakh/institution/year; target 100 B-schools
- Zetheta Fixed Income Certification: INR 7,999/candidate upon completing all 7 levels

5-Year Revenue Projection:

- Year 1: 8 banks × 40 users × INR 12,000 + 500 retail = INR 63 lakh
- Year 2: 20 banks + 2,000 retail + 5 universities = INR 2.8 crore
- Year 3: 35 banks + 8,000 retail + 20 universities = INR 8 crore
- Year 4: 45 banks + 20,000 retail + 50 universities + certification = INR 18 crore
- Year 5: Full market + 50,000 retail + 100 universities = INR 35 crore

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B3.4 Additional Sample Scenarios

Scenario 2: The Curve Inversion

Date: February 15. Overnight, the US Treasury 2Y-10Y curve inverts for the first time in 18 months. The Indian G-Sec curve has not inverted yet, but the 2Y-10Y spread has compressed from 80bps to 15bps in the past month. Your portfolio has 40% allocation to 10Y+ G-Secs and 25% in 1-3Y corporate bonds.

- Challenge 1: What is your current portfolio exposure to further curve flattening? Calculate using KRD decomposition.
- Challenge 2: If the Indian curve also inverts (2Y-10Y spread goes to -20bps), estimate the P&L impact on your portfolio using KRD vectors.
- Challenge 3: The CIO asks whether you should switch from bullet (concentrated 7Y) to barbell (2Y + 30Y). Calculate the convexity difference between the two strategies.
- Challenge 4: Run 500 Monte Carlo scenarios with elevated twist volatility (10bps vs normal 5bps). What is the VaR at 95% confidence?
- Scoring: KRD accuracy (60 pts), P&L estimation (40 pts), Strategy recommendation (30 pts), MC execution (20 pts)

Scenario 3: The Credit Downgrade Cascade

Date: September 8. A major Indian NBFC (rated AA+) is downgraded to A+ by CRISIL. Your portfolio holds INR 150 crore face value in this issuer's bonds (3Y and 5Y). The market reprices the spread from 85bps to 220bps overnight. Three other NBFC issuers in your portfolio (rated AA and AA-) see their spreads widen by 30-50bps in sympathy.

- Challenge 1: Recalculate the modified duration and convexity of the downgraded bonds at the new wider spread/yield.
- Challenge 2: The bond price dropped from 102.50 to 96.80. Compare the actual price change to your duration-convexity estimate. Why is there a discrepancy? (Answer: duration/convexity capture interest rate sensitivity, not credit spread sensitivity – need spread duration)
- Challenge 3: Recalculate portfolio VaR incorporating the increased spread volatility for NBFC sector.
- Challenge 4: The risk committee asks if you should sell or hold. Present a quantitative analysis showing P&L scenarios for hold vs sell at current price.

- Scoring: Repricing accuracy (50 pts), Spread vs rate decomposition (40 pts), VaR update (30 pts), Committee presentation (30 pts)

Scenario 4: The Operation Twist

Date: June 12. The RBI announces Operation Twist: simultaneously buying INR 10,000 crore of 10Y+ G-Secs and selling an equal amount of 1-3Y G-Secs. This is designed to flatten the curve. Your portfolio is long duration in the 10Y+ segment and short duration in the 1-3Y segment.

- Challenge 1: Estimate the yield impact using historical OT data (typically 10-15bps compression in 10Y, 5-8bps increase in 1-3Y).
- Challenge 2: Calculate the P&L impact on your portfolio using KRD for each affected bucket.
- Challenge 3: Should you pre-position ahead of further OTs? Quantify the risk-reward of increasing 10Y+ allocation by 10%.
- Challenge 4: What is the convexity impact? OT compresses long-end yields, which increases the price of long-duration bonds more than duration predicts (positive convexity).

Scenario 5: The Perpetual Bond Crisis

Date: March 3. An Indian private bank has issued AT1 perpetual bonds held in your portfolio (INR 50 crore face). The bank reports unexpected losses, and CRISIL puts the AT1 bonds on 'Rating Watch Negative'. The market yield spikes from 9.5% to 13.0% overnight.

- Challenge 1: Perpetual bond duration = (1+y)/y. Calculate duration at original yield (9.5%) and new yield (13.0%). Note how duration decreased from 11.5 to 8.7 years.
- Challenge 2: Perpetual bond convexity = $2/y^2$. Calculate convexity at both yields. Note the sharp change.
- Challenge 3: Estimate the price change using duration+convexity for a 350bps yield increase. Compared to actual market price. Discuss why the approximation may be less accurate for such a large move.
- Challenge 4: The risk committee wants to know the maximum possible loss if the yield goes to 20% (near-default). Calculate and present.

PART C: CASE STUDIES & REAL-WORLD APPLICATIONS

SECTION C1: RBI RATE HIKE CYCLE 2022-2023 – DURATION RISK CRYSTALLISED

C1.1 Background

Between May 2022 and February 2023, the Reserve Bank of India raised the repo rate by 250 basis points (from 4.00% to 6.50%) in the fastest tightening cycle in over a decade. The rate hikes came in rapid succession: 40bps emergency hike in May 2022, followed by 50bps each in June, August, and September, then 35bps in December 2022 and 25bps in February 2023.

This caused significant mark-to-market losses across all fixed income portfolios. Long-duration gilt funds lost 3-8% in NAV. Dynamic bond funds that were positioned for rate cuts suffered the most. Even liquid funds saw brief NAV dips during the May 2022 surprise.

C1.2 Duration And Convexity Analysis

Observation 1: Duration Predicted The Direction But Not The Magnitude

A gilt fund with portfolio modified duration of 8.0 years would predict a price decline of 8.0 × 2.50% = 20.0% for a 250bps rate increase. However, the actual loss was 15-18% because:

- Convexity partially offset the duration loss: $0.5 \times 120 \times (0.025)^2 = 3.75\%$ gain from convexity
- The curve did not shift in parallel – long-end rose less than the repo rate increase
- Some bonds rolled down the curve during the 10-month period, partially offsetting mark-to-market losses with carry

$$\text{Estimated Loss} = -\text{ModDur} \times \Delta y + 0.5 \times \text{Conv} \times (\Delta y)^2$$
$$= -8.0 \times 0.025 + 0.5 \times 120 \times 0.000625$$
$$= -20.0\% + 3.75\% = -16.25\%$$

Actual fund loss: approximately -16% to -18% (close to estimate!)

Observation 2: Non-Parallel Shift Caught Krd-Unaware Managers

The yield curve did not shift uniformly. The repo rate (overnight) rose 250bps, but the 10-year benchmark G-Sec yield rose only about 150bps (from 6.8% to 8.3% peak). The curve flattened significantly. Managers who hedged using only overall portfolio duration and assumed a parallel shift were still exposed to curve risk:

- Short-duration buckets (0-2Y): +200 to +250bps
- Medium-duration buckets (3-5Y): +150 to +180bps
- Long-duration buckets (10Y+): +100 to +150bps

A portfolio concentrated in the short end lost more than a duration-matched portfolio concentrated in the long end, despite having the same overall modified duration.

Observation 3: Callable Corporate Bonds Exhibited Negative Convexity

Corporate bonds with embedded call options showed negative convexity behaviour during the rate hike cycle. As yields rose above the coupon rate, the probability of the issuer exercising the call dropped to near zero, effectively extending the bond's duration beyond what the stated maturity suggested. This meant these bonds fell more in price than their stated modified duration predicted.

Observation 4: The Convexity Premium In Long-Duration Bonds

Investors holding 30-year G-Secs benefited from significant convexity during the subsequent rate cuts in 2024. The same convexity that slightly cushioned losses during the hike cycle provided outsized gains when rates started declining. A 50bps rate cut generated approximately 7.1% return for a 30Y bond with ModDur=14.2 and Conv=285:

$$\text{Return} = 14.2 \times 0.005 + 0.5 \times 285 \times (0.005)^2 = 7.10\% + 0.36\% = 7.46\%$$

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C1.3 Fund Performance Analysis

Fund Category	ModDur	Convexity	Predicted Loss	Actual Loss	Convexity Benefit
Liquid Fund	0.3	0.2	-0.08%	-0.05%	Negligible
Short Duration	2.5	9.0	-6.25%	-5.8%	0.28%
Medium Duration	3.8	22.0	-9.50%	-8.4%	0.69%
Dynamic Bond	6.0	55.0	-15.00%	-12.6%	1.72%
Gilt Fund	8.0	120.0	-20.00%	-16.3%	3.75%
10Y Constant Dur	10.0	150.0	-25.00%	-20.3%	4.69%

Note: The 'Convexity Benefit' column shows the percentage points of loss cushioning provided by convexity. For long-duration funds, this cushion exceeded 3-4 percentage points – a material difference.

C1.4 Simulation Scenario For Bond Risk Lab

Recreate the May 2022 – February 2023 rate cycle in compressed time (6 simulated business days, each representing one RBI policy action). Students must:

1. Compute portfolio duration and convexity before the first hike
2. After each hike, recalculate duration (it changes as yields change!) and estimate the cumulative P&L
3. Apply non-parallel shift: short rates up more than long rates for each hike
4. At day 4, recommend a portfolio adjustment to reduce duration exposure
5. At day 6, calculate the total P&L including convexity adjustment and compare to duration-only estimate
6. Final deliverable: Risk report showing that convexity saved the portfolio 3-5% of losses vs a convexity-ignorant estimate

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SECTION C2: US TREASURY MARKET – SILICON VALLEY BANK COLLAPSE (MARCH 2023)

C2.1 Background

On March 10, 2023, Silicon Valley Bank (SVB) was closed by the FDIC after a classic bank run triggered by unrealised losses in its bond portfolio. SVB had invested $91 billion in a held-to-maturity (HTM) bond portfolio, predominantly in US Treasuries and Agency MBS, with an average duration estimated at 5.6 years and significant concentration in the 10Y+ segment.

When the Federal Reserve raised rates by 525bps between March 2022 and July 2023, SVB's bond portfolio accumulated $17.7 billion in unrealised losses – exceeding the bank's $16.3 billion in total equity. When depositors became aware, a $42 billion bank run ensued in a single day.

C2.2 The Duration Risk Lesson

Quantifying Svb's Duration Exposure:

```
Portfolio: $91 billion HTM bonds
Estimated Modified Duration: 5.6 years
Rate increase: 525 basis points (0.0525)

Duration-estimated loss = 5.6 × 0.0525 × $91B = $26.7 billion
Actual unrealised loss: $17.7 billion
Difference explained by: convexity offset, coupon income, and
    MBS prepayment-related duration extension (negative
convexity)
```

Why The Actual Loss Differed from Duration Estimate:

- Convexity offset: $91B × 0.5 × ~45 × $(0.0525)^2$ = $5.6 billion (positive convexity cushion for Treasuries)
- MBS negative convexity: Agency MBS in the portfolio exhibited negative convexity – as rates rose, prepayments slowed, extending duration beyond the initial estimate
- Net MBS convexity loss: Estimated additional $2-3 billion of losses beyond duration estimate
- Total accounting: $26.7B (duration) - $5.6B (Treasury convexity) + $2.5B (MBS neg convexity) ≈ $23.6B – closer to actual after accounting for coupon income

C2.3 The Convexity Dimension – Mbs Negative Convexity

SVB's portfolio contained a significant allocation to Agency MBS (mortgage-backed securities). Unlike Treasury bonds which have positive convexity, MBS exhibit negative convexity due to the embedded prepayment option held by borrowers:

- When rates fall: Borrowers refinance (prepay), shortening MBS duration – the bond doesn't gain as much as positive convexity would predict
- When rates rise: Borrowers stop prepaying, extending MBS duration – the bond loses more than positive convexity would predict
- This 'heads I lose, tails I don't win as much' profile is negative convexity

For SVB, the effective duration of MBS extended from an estimated 4.5 years to over 7 years as rates rose – amplifying losses by roughly 50% beyond the initial duration estimate.

C2.4 Lessons For Fixed Income Risk Managers

- Lesson 1: Duration must be monitored dynamically. Modified duration changes as yields change – it is NOT a static number.
- Lesson 2: Convexity matters enormously for MBS portfolios. Positive convexity of Treasuries partially cushioned losses, but negative convexity of MBS amplified them.
- Lesson 3: HTM accounting does not eliminate risk – it only hides it from the income statement. The economic loss exists regardless of accounting classification.
- Lesson 4: Duration hedging using interest rate swaps would have cost SVB approximately $200-300 million annually in swap carry – trivial compared to the $17.7 billion loss and complete bank failure.

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SECTION C3: INDIAN GILT FUND RALLY 2019-2020 – CONVEXITY AS ALPHA DRIVER

C3.1 Background

Between February 2019 and May 2020, the RBI cut the repo rate by 250 basis points (from 6.50% to 4.00%). This coincided with the COVID-19 pandemic response. Long-duration gilt funds with modified duration of 8-12 years generated returns of 12-18% – far exceeding their coupon yields of 6-7%.

C3.2 The Convexity Bonus Quantified

The convexity effect was material for long-duration portfolios:

Fund Category	ModDur	Conv	Duration Gain	Convexity Bonus	Total Return
Short Duration	2.5	9	6.25%	0.28%	6.53% + carry
Gilt Fund (10Y focus)	8.0	120	20.00%	3.75%	23.75% + carry
Gilt Fund (Long)	10.0	150	25.00%	4.69%	29.69% + carry
10Y Constant Dur	10.0	150	25.00%	4.69%	29.69% + carry
Hypothetical 30Y Only	14.0	285	35.00%	8.91%	43.91% + carry

The convexity bonus contributed 3-9 percentage points of additional return beyond what duration alone predicted. For a hypothetical 30Y-only portfolio, the convexity bonus was nearly 9% – demonstrating why convexity is not just a risk measure but an alpha driver.

C3.3 Key Insight: Asymmetry Of Convexity

The most important practical lesson from comparing the 2022-23 rate hike cycle with the 2019-20 rate cut cycle: for the same 250bps move, positive convexity caused the portfolio to gain more on the way down than it lost on the way up. This asymmetry is the fundamental reason why portfolio managers pay a premium for bonds with higher convexity (accepting lower yields in exchange for better convexity profile).

SECTION C4: TATA SONS PERPETUAL BONDS – EXTREME DURATION AND CONVEXITY

C4.1 Background

Perpetual bonds (including AT1 bonds issued by Indian banks and perpetual NCDs issued by corporates like Tata Sons) present unique duration and convexity challenges. With no maturity date, their Macaulay Duration converges to (1+y)/y, and their convexity to $2/y^2$.

Duration and Convexity of Perpetuals at Different Yield Levels:

Yield (y)	MacD = (1+y)/y	ModD = 1/y	Convexity = $2/y^2$	Conv Impact @100bps
6%	17.67 yrs	16.67	555.56	+2.78%
8%	13.50 yrs	12.50	312.50	+1.56%
10%	11.00 yrs	10.00	200.00	+1.00%
12%	9.33 yrs	8.33	138.89	+0.69%
15%	7.67 yrs	6.67	88.89	+0.44%
20%	6.00 yrs	5.00	50.00	+0.25%

Key observation: As yields increase, both duration and convexity decrease for perpetuals. This means perpetuals become less sensitive to further yield increases – a natural stabiliser. Conversely, as yields fall, perpetual duration increases, amplifying gains.

C4.2 Lesson For Bond Risk Lab

Perpetual bonds require special treatment in duration-convexity models. The standard maturity-based approach does not apply. Bond Risk Lab includes a dedicated perpetual bond module where students must correctly calculate duration = (1+y)/y and convexity = $2/y^2$, recognising that these are yield-dependent and change significantly as spreads move.

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PART D: 15-DAY PROJECT METHODOLOGY & DELIVERABLES

SECTION D1: OVERVIEW

The 15-day project is structured as a progressive deep-dive into fixed income risk analytics. The intern moves from foundational bond mathematics through Monte Carlo simulation, machine learning, and Power BI dashboard deployment to produce a comprehensive submission package. Each phase builds on the previous. The methodology uses AI-accelerated development – interns use Claude.ai, ChatGPT, and Cursor for code generation. However, all calculations must be validated against analytical benchmarks.

SECTION D2: DAY-BY-DAY BREAKDOWN

Day 1 – Domain Orientation:

Learning Goals: Build familiarity with fixed income risk management and the Indian G-Sec market.

- Read Frank Fabozzi chapters 4-6 (Duration, Convexity, and Interest Rate Risk) – or equivalent AI-summarised content
- Visit CCIL website (ccilindia.com) – download 30 days of daily G-Sec yield data for the benchmark 10Y bond
- Visit FIMMDA website (fimmda.org) – study the yield curve construction methodology
- Download RBI's latest monetary policy statement – identify the current repo rate, stance, and inflation outlook
- Map the standard 11 key rate tenor points used by Indian bank Treasury/ALM desks (3M to 30Y)
- Identify the top 3 gilt mutual funds by AUM from AMFI data; note their published modified duration

Deliverable for Day 1: One-page summary of India's fixed income market with current yield curve snapshot, RBI policy stance, and benchmark duration/convexity for 3 schemes.

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Day 2 – Bond Pricing From First Principles:

Learning Goals: Implement all core bond mathematics in Python.

- Write bond_price(face, coupon_rate, ytm, years, freq) function
- Write macaulay_duration() function with full cash flow discounting
- Write modified_duration() function
- Write convexity() function
- Write dv01() function
- Validate all 5 functions against Excel PRICE(), DURATION(), MDURATION() for these test bonds: 7.26% GOI 2033 (YTM 7.35%), 91-day T-Bill (YTM 6.5%), 8.5% AAA Corp 2027 (YTM 8.2%)
- Compute duration and convexity for all 300 bonds in the provided portfolio CSV
- Generate summary statistics: min/max/mean/median for duration, convexity, DV01 by rating and sector

Deliverable for Day 2: Python script (bond_pricing.py) with 5 functions + validation against Excel. Summary table of portfolio duration/convexity statistics.

Day 3 – Portfolio Analytics:

Learning Goals: Aggregate bond-level metrics to portfolio level.

- Compute market-value-weighted portfolio Modified Duration
- Compute market-value-weighted portfolio Convexity
- Compute total portfolio DV01 (sum of individual DV01 × quantity)
- Build duration contribution breakdown by: credit rating (SOV/AAA/AA/A), sector, currency, key rate bucket
- Build price impact calculator: given Δy, compute portfolio P&L = -ModDur×Δy×MV + 0.5×Conv×$(\Delta y)^2$×MV
- Create visualisations: duration contribution pie chart, key rate duration bar chart, DV01 by currency
- Cross-validate: sum of KRD contributions ≈ total Modified Duration (within 1% tolerance)

Deliverable for Day 3: portfolio_analytics.py + 4 charts + summary table showing portfolio ModDur, Convexity, DV01.

Day 4 – Key Rate Duration:

Learning Goals: Decompose portfolio risk across the yield curve.

- Implement key_rate_duration() function: bump yield at each of 10 tenor points by 1bp, measure price change
- Compute KRD vector for full portfolio across 10 buckets: 3M, 6M, 1Y, 2Y, 3Y, 5Y, 7Y, 10Y, 20Y, 30Y
- Verify: sum of KRDs ≈ portfolio Modified Duration
- Analyse: which buckets contribute most to risk? Is the portfolio a barbell, bullet, or ladder?
- Build KRD heatmap visualisation showing contribution by rating × tenor bucket
- Scenario test: apply a steepening shock (short up 50bps, long up 25bps) and compute P&L using KRD

Deliverable for Day 4: key_rate_duration.py + KRD heatmap + steepening scenario analysis.

Day 5 – Monte Carlo Simulation (Parallel Shifts):

Learning Goals: Build MC engine for interest rate scenario generation.

- Implement BondRiskMC class with generate_scenarios() and compute_pnl() methods
- Generate 1,000 parallel yield shift scenarios: N(0, σ) where σ = 100bps annual volatility
- Compute scenario P&L using duration-convexity approximation for each scenario
- Calculate: VaR 95%, VaR 99%, CVaR 95%, CVaR 99%, maximum loss, maximum gain
- Plot: P&L distribution histogram with VaR lines marked, QQ plot to assess normality, P&L scatter vs yield change
- Compare: duration-only P&L vs duration+convexity P&L for extreme scenarios (>150bps shift)

Deliverable for Day 5: monte_carlo.py + P&L distribution chart + VaR summary table + duration vs convexity comparison.

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Day 6 – Monte Carlo (Non-Parallel Shifts):

Learning Goals: Extend MC to multi-factor yield curve scenarios.

- Add twist (slope) and butterfly (curvature) factors to MC engine
- Implement correlation structure: parallel-twist correlation = -0.3, parallel-butterfly = 0.1, twist-butterfly = 0.2
- Use Cholesky decomposition to generate correlated random factors
- Run 1,000 three-factor scenarios; compute scenario P&L using KRD for each tenor bucket
- Compare: VaR from parallel-only model vs 3-factor model (expect 3-factor to show different/higher risk)
- Sensitivity analysis: how does VaR change as twist volatility increases from 3bps to 10bps?

Deliverable for Day 6: Enhanced monte_carlo.py + 3-factor P&L distribution + VaR comparison table + sensitivity chart.

Day 7 – Case Study Research:

Learning Goals: Apply duration-convexity analytical lens to real-world events.

- Research RBI 2022-23 rate hike cycle: quantify duration and convexity impacts on gilt funds using actual performance data from AMFI
- Research SVB collapse: reconstruct the duration exposure and explain how negative convexity of MBS amplified losses
- Research Indian gilt fund rally 2019-20: quantify the convexity bonus that long-duration funds captured
- For each case: provide specific numbers (ModDur, Convexity, Δy, estimated vs actual P&L)

Deliverable for Day 7: Three case study analyses (6 pages total) with quantitative duration-convexity decomposition.

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Day 8 – Machine Learning Models:

Learning Goals: Train ML models to predict bond prices and compare to analytical methods.

- Feature engineering: create training dataset with coupon_rate, maturity, yield_rate, duration, convexity as features; price as target
- Split 80/20 train/test; standardise features
- Train Random Forest (100 trees), XGBoost (200 rounds, max_depth=4), Neural Network (3 layers: 128-64-32)
- Evaluate: RMSE, MAE, R^2 for each model on test set
- Feature importance: plot for RF and XGBoost; interpret – which features dominate?
- Implement SHAP for best model: summary plot, dependence plots for yield and maturity

Deliverable for Day 8: ml_models.py + model comparison table + feature importance chart + SHAP plots.

Day 9 – ML Sensitivity Analysis:

Learning Goals: Use ML for rapid scenario evaluation and compare it to analytical.

- For each of 50 yield perturbations (-200bps to +200bps in 8bps steps): predict prices using all 3 ML models
- For the same perturbations: compute analytical price change using duration+convexity
- Plot: ML-predicted price change vs analytical price change on the same chart. Where do they diverge?
- Analysis: ML captures non-linear effects beyond second-order – at what yield change does ML significantly outperform duration+convexity?
- Ensemble: create weighted average prediction from 3 models; does ensemble outperform individual models?

Deliverable for Day 9: sensitivity_analysis.py + ML vs analytical comparison chart + analysis document.

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Day 10 – Yield Curve Modelling:

Learning Goals: Fit parametric yield curve models to Indian data.

- Download 24 months of INR yield curve data from CCIL/FIMMDA (or use provided CSV)
- Implement Nelson-Siegel model fitting using scipy.optimize.minimize()
- Implement Svensson extension with 6 parameters
- Implement zero-coupon curve bootstrapping from G-Sec coupon bond prices
- Compare fit quality: R^2, RMSE, visual overlay of fitted vs actual for each model
- Compute forward rates from zero curve: 1Y-2Y forward, 2Y-5Y forward, 5Y-10Y forward
- Interpret: is the current curve predicting rate cuts or hikes? What does the forward curve imply?

Deliverable for Day 10: yield_curve.py + fitted curve charts + forward rate table + economic interpretation.

Day 11 – R Implementation:

- Implement complete R version: GBM, XGBoost, H2O Neural Network for duration and convexity prediction
- Create ggplot2 visualisations: sensitivity chart (dual axis for duration and convexity), VaR histogram
- Sensitivity analysis in R: adjusted duration and convexity across -2% to +2% rate change
- Export all results to CSV for Power BI import

Deliverable for Day 11: Complete R codebase (3 files) + ggplot2 outputs + CSV exports.

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Day 12 – Power BI Dashboards:

- Import all data (portfolio, yield curve, MC scenarios, ML results) into Power BI
- Build star schema data model with proper relationships
- Create 8+ dashboard pages: (1) Executive Summary with KPI cards, (2) Duration Contribution breakdown, (3) KRD Heatmap, (4) Yield Curve Evolution, (5) Monte Carlo P&L Distribution, (6) VaR/CVaR Monitor, (7) ML Model Comparison, (8) Scenario Analysis with What-If sliders
- Implement 40+ DAX measures covering portfolio analytics, VaR, contribution analysis, sensitivity
- Add What-If parameter for yield change: slider from -300bps to +300bps with real-time P&L update
- Add bookmarks for guided narrative walkthrough

Deliverable for Day 12: ConvexitySensitivity.pbix with 8+ pages and 40+ measures.

Day 13 – Gamification Concept Design:

- Write 3-page Bond Risk Lab business concept note: problem, solution, target audience, differentiation
- Design complete 1000-point scoring rubric (6 categories as defined in Part B)
- Create catalogue of 20 distinct scenarios with difficulty levels and learning objectives
- Design 10 achievement badges with specific criteria and descriptions
- Draft 5-year revenue projection with 3 pricing tiers

Deliverable for Day 13: Bond Risk Lab design document (8 pages).

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Day 14 – Validation And Polish:

- Validate all Python/R duration/convexity calculations against Excel crosscheck workbook
- Run DAX Studio performance profiling on all Power BI measures (target: <100ms each)
- Proofread all documentation; verify every formula
- Cross-check VaR numbers: parametric VaR vs MC VaR (should be within 5%)
- Ensure all GitHub files are committed with proper README

Deliverable for Day 14: Validated, polished draft with crosscheck evidence.

Day 15 – Final Submission And Handover:

- Compile all outputs into final report document (target: 60+ pages covering all 6 parts)
- Prepare 10-slide final presentation: Executive Summary (1), Market Overview (1), Duration/Convexity Framework (1), Monte Carlo & VaR (1), ML Models (1), Gamification Concept (2), Case Studies (1), Business Model (1), Key Learnings (1)
- Upload all code to GitHub; verify repository is PRIVATE
- Transfer GitHub repository ownership to @ZethetaIntern
- Deliver final presentation to evaluation panel

Deliverable for Day 15: Final PDF report + 10-slide presentation + all code files + repository transfer confirmation.

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SECTION D3: MANDATORY DELIVERABLES CHECKLIST

Deliverable 1: Main Report

Format: PDF or DOCX. Minimum 60 pages. Must contain all 6 parts (A through F) with professional formatting, original analysis, quantitative examples with specific numbers, and proper cross-references between sections.

- Part A: Training material covering all duration/convexity concepts with Python implementations
- Part B: Bond Risk Lab gamification concept with scoring rubric, scenarios, and progression system
- Part C: 3+ case studies with quantitative duration-convexity decomposition (not just narrative)
- Part D: Day-by-day project journal with actual outputs for each day
- Part E: Tools used with specific function references
- Part F: Self-assessment against the rubric with evidence for each criterion

Deliverable 2: Python Codebase

Format: .py files in GitHub repository. All files must run without errors. No hard-coded file paths.

- bond_pricing.py: bond_price(), macaulay_duration(), modified_duration(), convexity(), dv01() – validated against Excel
- portfolio_analytics.py: Portfolio-level aggregation, KRD computation, contribution analysis
- monte_carlo.py: BondRiskMC class with 3-factor scenarios, VaR/CVaR computation
- ml_models.py: Random Forest, XGBoost, Neural Network training and evaluation
- sensitivity_analysis.py: ML-based and analytical sensitivity comparison
- yield_curve.py: Nelson-Siegel, Svensson fitting; zero-coupon bootstrapping
- shap_explainability.py: SHAP analysis for best ML model
- var_backtest.py: VaR backtesting with Kupiec POF test

Deliverable 3: R Codebase

Format: .R files in GitHub repository.

- duration_convexity_analysis.R: GBM, XGBoost, H2O models for duration and convexity
- sensitivity_visualization.R: ggplot2 sensitivity charts
- sensitivity_results.csv: Exported results table

Deliverable 4: Power Bi Dashboard

Format: .pbix file. Must open without errors in Power BI Desktop.

- 8+ interactive pages with proper navigation
- 40+ DAX measures documented in measure dictionary
- What-If parameter for yield change (-300bps to +300bps)
- Bookmarks for guided narrative
- All measures profiled in DAX Studio to <100ms

Deliverable 5: Excel Validation Workbook

Format: .xlsx file. All cells must be formula-driven (no hard-coded values).

- Tab 1: Bond pricing validation for 5 test bonds (Python vs Excel comparison)
- Tab 2: Duration and convexity validation
- Tab 3: Portfolio-level aggregation crosscheck
- Tab 4: VaR validation (parametric vs MC comparison)
- Tab 5: Nelson-Siegel fitted yield curve vs actual

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Deliverable 6: Final Presentation

Format: PDF or Google Slides. 10 slides, professional design.

- Slide 1: Executive Summary
- Slide 2: Market Overview and Yield Curve
- Slide 3: Duration and Convexity Framework
- Slide 4: Portfolio Analytics Results
- Slide 5: Monte Carlo VaR Results
- Slide 6: ML Model Comparison
- Slide 7: Bond Risk Lab Concept (Gamification)
- Slide 8: Case Study Highlights
- Slide 9: Business Model and Revenue Projection
- Slide 10: Key Learnings and Next Steps

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SECTION D4: SUBMISSION INSTRUCTIONS

- All files named: 421_YourName_DeliverableName (e.g., 421_RahulKumar_MainReport.pdf)
- GitHub repository must be PRIVATE; transfer to @ZethetaIntern by 11:59 PM Day 15
- All PDFs must be readable and not password-protected
- Excel files in .xlsx format; Python files with .py extension
- No notebooks (.ipynb) – convert all to .py scripts
- All daily deliverables across Days 1–15 are to be submitted on **Day 15** as a consolidated final submission. Students should combine their work into **one or two PDFs and/or one Excel file** wherever possible, rather than submitting separate files for each day. Clearly label each section within the consolidated document by its respective day and topic for ease of assessment.

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PART E: TOOLS & PLATFORMS GUIDE

SECTION E1: PYTHON ENVIRONMENT

E1.1 Required Libraries

```
pip install numpy pandas scipy scikit-learn xgboost tensorflow
pip install matplotlib seaborn plotly shap
pip install openpyxl xlsxwriter  # For Excel export
```

Essential Python Functions For Fixed Income:

NumPy – Numerical Computing:

- np.random.normal(0, vol, n_scenarios): Generate normally-distributed yield changes for Monte Carlo
- np.percentile(pnl_array, 5): Compute VaR at 95% confidence
- np.linalg.cholesky(correlation_matrix): Cholesky decomposition for correlated factor generation
- np.power(1 + rate, periods): Compound interest calculations for bond pricing
- np.cumsum(): Cumulative sum for cumulative P&L tracking

SciPy – Optimisation and Statistics:

- scipy.optimize.minimize(): Fit Nelson-Siegel and Svensson yield curve models
- scipy.stats.norm.ppf(0.95): Z-score for parametric VaR (= 1.6449)
- scipy.stats.norm.pdf(): Probability density for P&L distribution fitting
- scipy.interpolate.CubicSpline(): Interpolate yield curve between tenor points

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scikit-learn – Machine Learning:

- RandomForestRegressor(n_estimators=100): Ensemble model for bond price prediction
- train_test_split(X, y, test_size=0.2): Split data for model validation
- mean_squared_error(): Compute RMSE for model evaluation
- StandardScaler(): Normalise features for neural network input
- cross_val_score(): K-fold cross-validation for robust evaluation

XGBoost:

- XGBRegressor(n_estimators=200, max_depth=4, learning_rate=0.1)
- Key parameters: subsample=0.8, colsample_bytree=0.8 (regularisation)
- early_stopping_rounds=10: Prevent overfitting

TensorFlow/Keras:

- Sequential model with Dense layers: architecture [128, 64, 32, 1]
- Dropout(0.2): Regularisation between layers
- Adam optimiser with learning_rate=0.001
- EarlyStopping callback: patience=10, restore_best_weights=True

SHAP:

- shap.TreeExplainer(model): For tree-based models (RF, XGBoost)
- shap.DeepExplainer(model, X_train): For neural networks
- shap.summary_plot(): Feature importance overview
- shap.dependence_plot('yield_rate', shap_values, X): Per-feature relationship

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Matplotlib/Seaborn – Visualisation:

- plt.hist(pnl, bins=50): P&L distribution histogram
- plt.axvline(var_95, color='red'): VaR threshold marker
- sns.heatmap(krd_matrix): Key Rate Duration heatmap by rating and tenor
- plt.plot(yield_changes, price_changes): Sensitivity curve

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SECTION E2: R ENVIRONMENT

E2.1 Required Packages

```
install.packages(c('tidyverse','caret','xgboost','Metrics','ggplot2'))
install.packages('h2o')  # Requires Java 8+
install.packages('quantmod')  # Financial data retrieval
```

Key R Functions:

- caret::train(): Unified interface for GBM, Random Forest, and many other ML models
- caret::trainControl(method='cv', number=5): 5-fold cross-validation
- xgboost::xgboost(): Direct XGBoost interface for maximum control
- h2o::h2o.deeplearning(): Deep neural network with dropout regularisation
- Metrics::rmse(): Root Mean Squared Error computation
- ggplot2::geom_line() + geom_ribbon(): Sensitivity plot with confidence bands

Sample ggplot2 Sensitivity Visualisation

```
ggplot(results, aes(x = RateChange)) +
    geom_line(aes(y = AdjDuration, color = 'Duration'), linewidth = 1.2) +
  geom_line(aes(y = AdjConvexity/10, color = 'Convexity/10'), linewidth = 1.2) +
  geom_hline(yintercept = 0, linetype = 'dashed') +
  scale_x_continuous(labels = scales::percent) +
  labs(title = 'Duration & Convexity Sensitivity to Interest Rate Changes',
       x = 'Yield Change', y = 'Adjusted Metric', color = 'Risk Measure') +
  theme_minimal(base_size = 14)
```

SECTION E3: POWER BI AND DAX

Power BI Desktop: Free download from Microsoft (https://www.microsoft.com/)

Power BI is the primary dashboard platform for this project. The .pbix file must contain:

- Star schema data model with Bonds (fact), DimRating, DimSector, DimCurrency, DimTenor, MCScenarios tables
- Relationships: all one-to-many from dimension to fact; single cross-filter direction
- 40+ DAX measures organised into measure groups: Portfolio Analytics, VaR, Contribution, Scenario, ML
- What-If parameter table for yield change slider (-300bps to +300bps in 10bps steps)

DAX Studio: Free download (https://daxstudio.org/)

DAX Studio is mandatory for performance profiling. Every DAX measure must be tested:

- Connect to Power BI Desktop instance via DAX Studio
- Execute each measure and capture Server Timings
- Target: <100ms for all measures; <500ms for complex iterators
- Identify bottlenecks: SE (Storage Engine) vs FE (Formula Engine) – SE queries should dominate
- Optimise: use variables (VAR/RETURN) to avoid repeated sub-expression evaluation

Tabular Editor: Free/paid versions (https://tabulareditor.com/)

- Use for bulk editing DAX measures (faster than Power BI Desktop UI)
- Best Practice Analyser: run built-in rules to check for common DAX anti-patterns
- Export model as .bim (JSON) file for version control in GitHub

SECTION E4: DATA SOURCES

Primary Data Sources for Indian Fixed Income:

Source	URL	Data Available	Update Frequency
CCIL	https://www.ccilindia.com/	G-Sec prices, yield curves, trade data	Daily
FIMMDA	https://fimmda.org/	Corporate bond valuation matrix	Daily
RBI	https://rbi.org.in/	Policy rates, auction data, SGL holdings	Weekly/Event
NSE	https://www.nseindia.com/	G-Sec benchmark yields, corporate bond trades	Daily
AMFI	https://www.amfiindia.com/	Mutual fund NAV data, AUM statistics	Daily
Bloomberg	https://www.bloomberg.com/	Real-time bond data (subscription)	Real-time

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SECTION E5: AI RESEARCH TOOLS

Permitted AI Tools and Usage Guidelines:

- Claude.ai: Primary tool for Python/R/DAX code generation, mathematical derivation, debugging, and concept explanation
- ChatGPT: Alternative for code generation and concept validation; use for second opinions on complex formulas
- Cursor AI: AI-powered code editor for rapid Python/R development; real-time code suggestions
- Perplexity.ai: Research synthesis for case studies and regulatory analysis; provides source citations
- GitHub Copilot: In-editor code completion for Python and R scripts

Mandatory Validation Requirements:

AI tools are encouraged for code generation and research acceleration. However, all outputs must be validated:

- Every bond pricing formula must be cross-checked against Excel PRICE()/DURATION() functions
- Every ML model must report RMSE on held-out test set
- Every VaR calculation must be compared to parametric VaR as a sanity check
- Every DAX measure must be profiled in DAX Studio for performance
- Case study facts (dates, numbers, regulatory references) must be verified against primary sources

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PART F: SUBMISSION PROTOCOL

Projects are assessed using a comprehensive 1000-point rubric covering seven dimensions: Problem Understanding, Solution Quality, Research & Analysis, Presentation & Clarity, Innovation & Creativity, Feasibility & Practicality and CV alignment. Assessment is conducted through a combination of manual review and AI-powered evaluation.

SECTION F1: AUTOMATIC DISQUALIFIERS

The following issues result in automatic project rejection requiring re-submission:

- Repository not transferred to ZethetaIntern by the deadline
- Personal API keys, passwords, or credentials committed to the repository
- Test coverage below 60% (hard minimum)
- Plagiarism: code copied from existing repositories without attribution and modification
- Data leakage: training data contaminating the test set (especially random splits on material data)
- Non-functional Docker build: docker-compose up must start all services without errors
- Missing technical report or README
- Personal forks or copies retained after transfer (verified by Zetheta's security team)

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SECTION F2: SUBMISSION FINAL CHECKLIST

Before executing the repository transfer, verify every item:

- All code committed and pushed to main branch (no dangling branches with unmerged work)
- All tests passing in CI/CD pipeline (green checkmark on latest commit)
- DVC remote configured and all data files pushed
- MLflow experiments exported and committed
- Docker build succeeds without errors (test with docker-compose up --build)
- README.md is complete with architecture diagram and screenshots
- Technical report is complete (15–20 pages, 20+ references)
- Sphinx API documentation generated and committed to docs/
- No personal API keys or credentials in repository (use .env.example template)
- LICENSE file present with Zetheta IP attribution
- .gitignore properly configured (no data files, no __pycache__, no .env, no mlruns/)
- Final commit message: "Final submission – MatRisk AI v1.0 — [Your Email Id] — [Date]"
- Git tag created: git tag -a v1.0 -m "MatRisk AI v1.0 Final Submission"
- Repository to be transferred through Zetheta task pane in the projects ideally – if not possible then Repository transferred via Settings → Danger Zone → Transfer Ownership → ZethetaIntern
- Confirmed repository appears under github.com/ZethetaIntern/matrisk-ai
- All local forks, copies, and clones deleted from personal devices
- Confirmation email sent to project supervisor with transfer timestamp and URL

Post-Transfer: ZethetaIntern will review the repository and score/ rank the submissions.

IP PROTECTION PROTOCOL:

All IP arising from this project including submissions made is the exclusive property of Zetheta Algorithms Pvt. Ltd. as per the confidentiality terms accepted at project commencement

- All innovations documented with dated lab notebook entries in the technical report and git commit history
- Novel algorithms include pseudo-code and mathematical formulation in patent-ready format
- Comparative results against prior art quantified and documented in the model performance report
- The GitHub repository commit history serves as a timestamped invention record under Indian patent law

ZeTheta
ADD EXPERIENCE TO KNOWLEDGE

END OF PROJECT DOCUMENT